VERA: Support-Aware Certification of Representation Edits Under Deployment Shift

Anonymous submission

Abstract

A representation edit may appear safe on IID validation data yet damage its target task or expose sensitive information after deployment reweighting. We introduce VERA, a decision layer that asks whether a registered edit should be deployed under a declared shift. VERA jointly audits paired target harm and a fresh heterogeneous attacker portfolio, then returns a simultaneous vector envelope of supported environment- and source-conditional density-ratio budgets, its limiting common radius, or ABSTAIN. We prove finite-sample validity throughout the reported envelope, derive a minimax prospective allocation of certification evidence for additive multi-cell contracts with a locally matching lower bound in compact independent-cell regimes, and prove an impossibility result for new deployment support. Finite-family testing follows Learn Then Test; the proposed contribution is the representation-intervention certificate that couples paired harm, balanced leakage, a continuum of groupwise shifts, and an explicit support boundary. In a prior IID uncertainty-control study, validation-only selection deployed contract-violating edits in 27.3% of 128 decisions versus 0.8% for the fixed-profile certificate; this result does not establish robustness for $\Gamma > 1$. The first preregistered controlled-shift primary made 38 VERA deployments with 0 violations but failed its usefulness gate because the 95% lower bound was 15.1%. We then locked an independent, non-pooling follow-up before seeds 109–172. At the higher preregistered 8,000-observation budget, validation selection violated 24/186 deployed decisions while VERA deployed 59 edits with 0 shifted-contract violations, retained 59/183 oracle-safe opportunities (32.2%, 95% CI [27.5%, 36.8%]), and passed every registered follow-up gate. In the earlier support-boundary study, VERA abstained on Camelyon17 because the deployment hospital was absent from certification support.

1 Introduction

Representation editing is an intervention. It changes what downstream models can recover from a frozen representation, often to remove a protected, nuisance, or source concept. INLP, R-LACE, LEACE, kernel methods, TaCo, SPLINCE, and MANCE++ provide increasingly expressive ways to construct such edits (Ravfogel et al. 2020, 2022a; Belrose et al. 2023; Ravfogel et al. 2022b; Jourdan et al. 2023; Holstege, Ravfogel, and Wouters 2025; Avitan, Goldberg, and Elazar 2026). Their successes do not make every resulting edit deployable. Removing a correlated concept can damage the target, a probe used by the eraser can understate leakage to a different attacker, and an IID validation result need not survive reweighting at deployment.

This paper asks a narrower, operational question: *should this representation edit be deployed under this declared shift?* VERA does not construct a new eraser. It certifies whether a representation edit can be deployed under a declared, supported reweighting shift while jointly controlling incremental target harm and recovery by a registered attacker portfolio. It evaluates the output of registered erasers and returns a deployment decision plus a sensitivity certificate: a vector shift envelope, a limiting common radius, the contract that limits that radius, and the support boundary beyond which it must abstain. The target audit is paired. For the same example, it subtracts the identity representation's zero-one loss from the edited representation's loss, isolating incremental harm rather than conflating it with baseline error. The shift class is stratified: within each environment, the target law has a bounded density ratio relative to its certification law. Leakage is the equal mean of two robust source-class recalls, each under a bounded source-conditional density ratio. Environment weights and source prevalence may therefore vary arbitrarily; ordinary prevalence-weighted accuracy is not the contract. At a declared profile, VERA uses a candidate-wise intersection–union test. Separately, it simultaneously upper-bounds every candidate-contract curve and inverts them into a support-aware shift envelope and common radius $\widehat{\Gamma}_e$. A zero radius means that even the IID contract is not established. New support forces abstention.

The statistical mechanism must be positioned carefully. Risk-controlling prediction sets, Learn Then Test (LTT), Conformal Risk Control, and Pareto Testing already calibrate finite algorithm families against one or several risks; recent work also couples risk, acceptance, and utility and provides group-conditional PAC control (Bates et al. 2021; Angelopoulos et al. 2025, 2024; Laufer-Goldshtein et al. 2023; Yu and Liu 2026; Huang et al. 2026). Prompt Risk Control applies related candidate-wise bounds to LLM prompts and includes an estimated covariate-shift correction (Zollo et al. 2024). Weighted risk control and robust validation already address important forms of covariate and distribution shift (Tibshirani et al. 2019; Almeida et al. 2025; Zecchin et al. 2025; Cauchois et al. 2024; Ai and Ren 2024). VERA uses these foundations. Its proposed contribution is the representation-intervention object they do not report: a simultaneous, support-aware envelope of groupwise reweighting budgets for paired edit harm and an entire retrained-attacker portfolio. The common shift radius is one summary of that envelope; unsupported required cells mark the boundary at which it cannot be identified.

We make four contributions. First, we define a support-aware vector certificate for paired representation interventions, coupling environment-conditional incremental target harm to source-prior-invariant leakage from freshly retrained attackers. Second, we give simultaneous finite-sample validity over registered edits, environments, attackers, source classes, and a continuum of declared reweighting budgets, then derive minimax prospective evidence allocation for additive multi-cell contracts and a locally matching lower bound for compact independent-cell designs. Third, an indistinguishability theorem shows why the certificate must collapse when deployment may use unseen support. Fourth, we conduct a preregistered cross-modal official-code study that asks whether this certificate changes deployment decisions, prevents unsafe deployments, and preserves useful safe opportunities.

The stakes are concrete. A hospital may erase acquisition site from pathology features, a hiring system may remove gender from occupational embeddings, a toxicity model may suppress identity mentions, or a gait model may remove recording site. In each case, the edit can reduce measured leakage while damaging the intended prediction or failing in a reweighted group. VERA turns that ambiguity into a falsifiable, support-scoped deployment statement.

2 Related Work

Concept erasure and probing. Linear erasure includes iterative nullspace projection, adversarial low-rank projection, and the closed-form LEACE solution (Ravfogel et al. 2020, 2022a; Belrose et al. 2023). Kernel erasure, rate-distortion objectives, unaligned-attribute removal, TaCo, SPLINCE, and MANCE extend the construction space (Ravfogel et al. 2022b; Basu Roy Chowdhury et al. 2023; Shao, Ziser, and Cohen 2023; Jourdan et al. 2023; Holstege, Ravfogel, and Wouters 2025; Avitan, Goldberg, and Elazar 2026). Nonlinear guardedness can remain after linear removal, while perfect erasure faces fundamental utility limits (Akbari, Afshari, and Boddeti 2025; Chowdhury et al. 2025). FARE and MMD-B-Fair provide important certificates for restricted fair-representation objectives (Jovanovic et al. 2023; Deka and Sutherland 2023). VERA neither replaces these erasers nor certifies absence of all information. It audits their candidate outputs against a fixed portfolio. That portfolio follows the broader lesson that a probe's result depends on its hypothesis class and training protocol (Hewitt and Liang 2019; Pimentel et al. 2020; Voita and Titov 2020; Elazar et al. 2021).

Risk control and abstention. LTT turns calibration into testing over a finite family, and Pareto Testing extends the view to multiple risks and objectives (Angelopoulos et al. 2025; Laufer-Goldshtein et al. 2023). Conformal Risk Control treats general monotone losses (Angelopoulos et al. 2024). Selective conformal risk control and joint adaptive certificates couple risk, acceptance, and utility at the example or prediction-set level, while group-conditional PAC methods control routing risk within registered groups (Xu, Guo, and Wei 2025; Yu and Liu 2026; Huang et al. 2026). These works invalidate any claim that VERA invents finite-family, joint, selective, or groupwise acceptance. Prompt Risk Control is a particularly close application template: it selects LLM prompts using high-probability bounds on mean, tail, and dispersion risks and corrects covariate shift using unlabeled target data (Zollo et al. 2024). VERA instead audits paired representation interventions without a target sample or estimated importance weights, reports worst-case guarantees over a declared bounded reweighting class, retrains a heterogeneous attacker portfolio, and makes unsupported cells an explicit impossibility boundary. Selective classification and SelectiveNet reject individual examples (El-Yaniv and Wiener 2010; Geifman and El-Yaniv 2017, 2019); VERA instead rejects an entire intervention and provides a predeployment, support-scoped sensitivity certificate.

Shift-robust evaluation. Known importance weights can restore conformal or high-probability risk control under a specified covariate shift (Tibshirani et al. 2019; Almeida et al. 2025; Zecchin et al. 2025). Robust validation and fine-grained robust conformal inference provide ambiguity-set guarantees (Cauchois et al. 2024; Ai and Ren 2024), while robust model evaluation and CVaR concentration supply closely related technical tools (Najafi, Sani, and Farnia 2025; Thomas and Learned-Miller 2019; Waudby-Smith and Ramdas 2024). Distributional fairness certificates and Wasserstein fairness audits are nearby applications (Kang et al. 2022; Ehyaei, Farnadi, and Samadi 2025). Worst-case subpopulation treatment effects are especially close to our paired-harm view (Jeong and Namkoong 2020). VERA does not claim these tools or generic robust fairness auditing. It applies bounded reweighting to a selected representation edit's paired target and retrained-attacker contracts and reports the admissible support-scoped envelope. Group DRO and robustness benchmarks motivate the observed-group view (Sagawa et al. 2020; Koh et al. 2021; Sohoni et al. 2020), and domain-adaptation impossibility results motivate explicit support assumptions (David et al. 2010; Subbaswamy, Adams, and Saria 2021).

3 Problem Setup

Let $Z \in \mathbb{R}^d$ be a frozen representation, Y its target label, and S a source or sensitive label. An edit $e \in \mathcal{E}$ is fitted without the certification fold and maps Z to Z^e. The identity is $e = 0$. Fresh target models are fitted after every edit. For certification example i, define paired target harm

$$H_{e,i} = \mathbf{1}\{f_e(Z_i^e) \neq Y_i\} - \mathbf{1}\{f_0(Z_i) \neq Y_i\} \in \{-1, 0, 1\}. \tag{1}$$

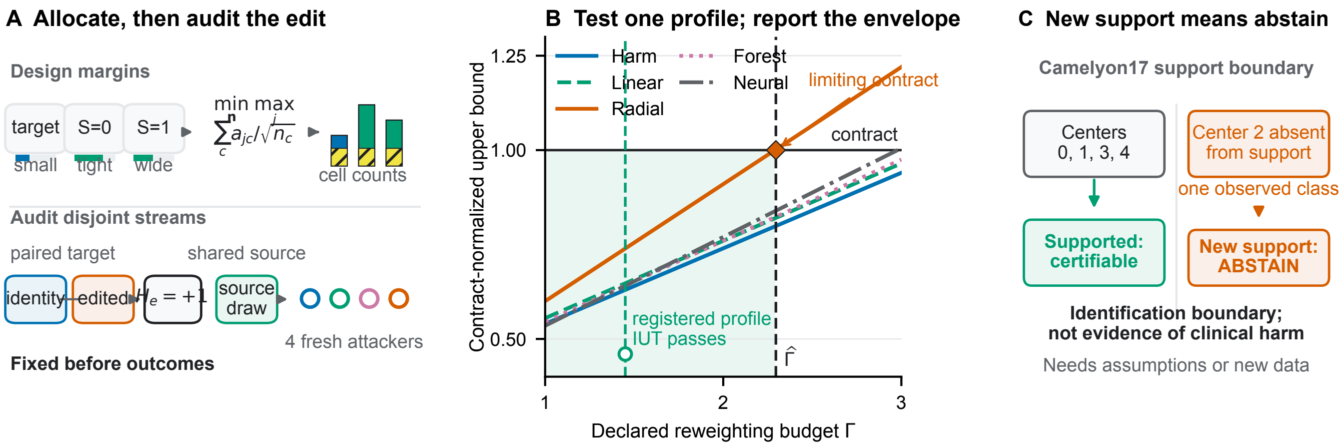

Figure 1: **VERA allocates evidence, audits one profile, and exposes its support boundary.** (A) Independent design-fold margins determine disjoint certification-stream counts before certification outcomes are observed. Paired identity/edit predictions measure target harm, while one source-class draw evaluates every registered fresh attacker. The primary controlled study uses the preregistered square-score allocator; the disjoint-stream additive comparison is a separately, prospectively specified secondary analysis. (B) The candidate IUT decides the registered profile, while simultaneous schematic upper-risk curves provide an inspectable envelope and limiting common radius. (C) The guarantee cannot cross new support: Camelyon17 certification centers are 0, 1, 3, and 4, whereas external center 2 is absent from that support, so VERA abstains without an additional cross-center assumption. This is an identification boundary, not evidence of clinical harm.

In the real protocol, identity is the paired comparator rather than a selectable erasure action. If a deployment protocol may select identity, it must register identity as one additional candidate in the finite-family correction; the separate family-size coverage grid does so. For each preregistered attacker $a \in \mathcal{A}$, let $L_{e,a,i} = \mathbf{1}\{a_e(Z_i^e) = S_i\} \in \{0,1\}$. Attackers are retrained after each edit using eraser-training data, but are fixed before certification outcomes are used.

For certification reference law P and $\Gamma \geq 1$, define

$$\begin{aligned} \mathcal{Q}_\Gamma(P) &= \left\{ Q \ll P : 0 \leq \frac{\mathrm{d}Q}{\mathrm{d}P} \leq \Gamma \quad P\text{-a.s.} \right\}, \\ \rho_{\Gamma,P}(V) &= \sup_{Q \in \mathcal{Q}_\Gamma(P)} \mathbb{E}_Q[V]. \end{aligned} \tag{2}$$

The target contract uses P_g, the certification law conditional on environment g. For binary source, leakage uses the balanced robust accuracy

$$B_{e,a}(\eta_0, \eta_1) = \frac{1}{2} \sum_{s \in \{0,1\}} \rho_{\eta_s, P_s}(L_{e,a}), \tag{3}$$

where P_s is source-conditional. Balanced accuracy itself is standard (Brodersen et al. 2010); our change is to robustify each recall and couple the resulting attacker contracts to paired edit harm. A constant attacker scores $1/2$, not one, and changing source prevalence does not change the estimand. At zero or unit prevalence this is a standardized two-conditional functional, not ordinary accuracy observable in an absent class. The population common representation-edit shift radius is

$$\begin{aligned} \Gamma_e^\star = \sup\{\Gamma \geq 1 :& \rho_{\Gamma,P_k}(V_{e,k}) \leq c_k \\ & \text{for every registered contract } k\}, \end{aligned} \tag{4}$$

with value zero if a contract fails at $\Gamma = 1$.

This radius is a sensitivity statement, not an estimate of how much shift will occur. A user declares a profile of plausible density-ratio caps; VERA states whether the available certification sample supports the contracts throughout that profile. Conditioning target harm on environment prevents a favorable environment mixture from hiding damage in a smaller group. Standardizing leakage to equal source-class weight similarly prevents source prevalence from making an attacker appear weak. These two conditionings need not describe one factorized joint model: the claim applies when each registered conditional law obeys its own stated cap.

4 VERA

The bounded-reweighting risk has the CVaR dual

$$\rho_{\Gamma,P}(V) = \inf_{\eta \in [a,b]} \{\eta + \Gamma \mathbb{E}_P[(V-\eta)_+]\}. \tag{5}$$

For general bounded variables, a Dvoretzky–Kiefer–Wolfowitz band gives a uniform upper curve. Our discrete audit variables admit tighter exact curves. For Bernoulli leakage with success probability p, $\rho_{\Gamma,P}(L) = \min(1, \Gamma p)$. For paired harm with probabilities (p_-, p_0, p_+) on $(-1, 0, 1)$, set $a = \Gamma p_+$ and $b = \Gamma(1 - p_-)$. The adversary places its permitted mass on $+1$, then zero, then -1:

$$\rho_{\Gamma,P}(H) = \begin{cases} 1, & a \geq 1, \\ a, & a < 1 \leq b, \\ a + b - 1, & b < 1. \end{cases} \tag{6}$$

VERA substitutes one-sided Clopper–Pearson bounds. For paired harm it uses an upper bound U_+ for p_+ and lower bound L_- for p_-, clips $\widetilde{p}_+ = \min(U_+, 1 - L_-)$ to preserve the simplex, and splits the component error between the two tails. Balanced leakage averages two class recall bounds, splitting its component error between the source classes.

At the fixed deployment profile, each candidate is an intersection–union test: all of its target and balanced-leakage components must pass at level $\delta/|\mathcal{E}|$. If a candidate is unsafe, false acceptance implies undercoverage of at least one fixed violated component. A union bound is needed over candidates, but not over components. The separately reported envelope uses Bonferroni coverage over every candidate–contract pair because every curve remains available for post-certification inspection.

Let $U^T_{e,g}(\gamma_g)$ denote the simultaneous target-harm upper curve in observed environment g, and let $U^L_{e,a,s}(\eta_s)$ denote attacker a's source-class recall curve. VERA's central reported object is

$$\widehat{\mathcal{E}}_e = \{(\boldsymbol{\gamma}, \boldsymbol{\eta}) : U^T_{e,g}(\gamma_g) \le \tau \quad \forall g, \\ \tfrac{1}{2}\textstyle\sum_s U^L_{e,a,s}(\eta_s) \le \lambda \quad \forall a\}. \tag{7}$$

Coordinates for unsupported required cells are zero. The common radius is the largest diagonal point $(\Gamma\mathbf{1}, \Gamma, \Gamma)$ in this envelope, right-censored at the registered computational cap. Reporting the vector envelope as well as its diagonal summary reveals which environment, source class, or attacker limits an edit.

Decision rule. Edits, learners, attackers, contracts, and the allocation rule are fixed without certification outcomes. An independent design fold chooses the max-min-margin edit for evidence allocation. Each edit then receives a fixed-profile IUT and a separate simultaneous envelope; VERA returns the registered utility-best accepted edit or ABSTAIN. Probability is over independent certification observations conditional on all construction and design objects. Candidate selection and later profile inspection remain valid because the coverage event already contains every registered candidate and curve. Integrating gives unconditional validity when construction is random but independent, and no target-deployment sample is required.

Theorem 1 (Fixed-profile VERA)**.** *Fix M candidate edits and all audit variables independently of certification. At one declared shift profile, accept a candidate only if every component upper bound clears its threshold at level δ/M. Then the probability of deploying any candidate that violates at least one declared target or balanced leakage contract is at most δ.*

For an unsafe candidate, fix one violated population component. Accepting that candidate implies undercoverage of this component; intersecting more tests cannot increase that probability. Union bounding only over the M candidates proves the result. This is the LTT-style finite-family decision used for the primary experiment.

LTT supplies finite-family testing, not VERA's population object. VERA couples paired target harm with source-balanced recovery by retrained attackers and inverts every conditional curve into a supported vector shift region. For example, Bernoulli leakage $p = 0.04$ passes an IID threshold 0.05, yet at $\Gamma = 2$ its robust risk $\min(1, \Gamma p) = 0.08$ fails; IID LTT certifies only the first statement. Scalar pooling also loses required coordinates: risks 0.2 and 1.2, normalized by their contracts, have mean 0.7 although one contract fails. VERA requires both to pass. One simultaneous distribution-function event covers every registered curve and Γ, so post-certification profile inspection needs no uncountable union bound. Controlled environment risks remain controlled under arbitrary mixtures, source prevalence is absent from balanced leakage, and the diagonal gives the common radius. Full proofs are in the supplement.

Theorem 2 (Support-aware envelope)**.** *With probability at least $1 - \delta$, simultaneously for every registered edit e, every profile in $\widehat{\mathcal{E}}_e$ satisfies all target-harm and balanced-leakage contracts. Environment mixture weights and source prevalence may be selected after certification without another multiplicity correction.*

The proof event covers every edit, environment, attacker, source class, and every $\Gamma \ge 1$. Conditional robust-risk bounds imply each component contract; arbitrary mixtures of controlled environment risks remain controlled by linearity, and source prevalence is absent from the standardized leakage functional. Intersecting or choosing objects already covered by the same event does not spend additional error probability. This argument establishes the deployment-law guarantee only for supported laws whose registered conditionals lie inside the declared envelope.

The same analysis quantifies when abstention can disappear. For a bounded component of range length R, declared budget Γ_0, and population margin $m > 0$ below its threshold, a DKW construction passes that component with power at least $1 - \beta/J$ whenever

$$n \ge \frac{\Gamma_0^2 R^2}{2m^2}\left(\sqrt{\log\frac{2}{\alpha}} + \sqrt{\log\frac{2J}{\beta}}\right)^2. \tag{8}$$

For additive contracts, let a_{jc} be cell c's normalized DKW coefficient in contract j. VERA solves the convex program $\min_{\boldsymbol{n}} \max_j \sum_c a_{jc}/\sqrt{n_c}$ under the evidence budget and cell floors. With one contract and inactive floors, $n_c \propto a_{1c}^{2/3}$; with one cell per contract it reduces to squared-score allocation. Margins come from the independent design fold, so this optimizes the chosen edit's power surrogate while preserving validity for every edit. Conversely, distinguishing an all-safe independent Bernoulli vector from every one-unsafe-component alternative requires $N = \Omega(\log(1/(\alpha + \beta))\sum_j \Gamma_j^2/m_j^2)$ under interior probabilities, locally matching the upper bound's shift and margin dependence. The supplement gives the exact-KL form without that interior restriction.

Theorem 3 (Unsupported-cell impossibility)**.** *Suppose a required audit cell has zero certification probability, and the model class contains safe and unsafe worlds that induce identical protocol observations but disagree on the contract in that cell. If deployment may concentrate there, any certification-data-only protocol with uniform false-acceptance control δ accepts the edit in the safe world with probability at most δ.*

The proof couples safe and unsafe worlds with identical observations, so any randomized protocol accepts equally often in both; false-acceptance control in the unsafe world limits safe-world acceptance to δ. This covers unseen environments and missing environment–source-label cells. Camelyon17 center 2 and its single observed source class therefore trigger abstention, not a claim that its measured target outcome violates a contract. Only new support or an explicit structural assumption removes this identification boundary.

5 Experimental Protocol

The central controlled supported-shift study was preregistered before seeds 45–108. It crosses five erasers (INLP, R-LACE, LEACE, TaCo, and MANCE++), four supported datasets, and 64 fresh seeds: 1,280 dataset–method runs and 3,072 candidate evaluations. Waterbirds, CivilComments, Bios, and GaitPDB span vision, text, and time series (Sagawa et al. 2020; Koh et al. 2021; Borkan et al. 2019; De-Arteaga et al. 2019). The prior 32-seed IID study is used only for uncertainty control. Camelyon17 is excluded from supported-shift endpoints and retained as the unseen-hospital support boundary. Full targets, sources, and support scopes appear in the supplement.

The official frontier contains 12 edits per dataset–seed block: INLP ranks $\{1, 2, 4, 8\}$, R-LACE ranks $\{1, 4\}$, closed-form LEACE, TaCo removal counts $\{1, 2, 3, 5\}$, and one MANCE++ configuration. Each candidate is generated through a pinned official upstream entry point under the common frozen-feature protocol; this does not reproduce every upstream paper's full training pipeline. No proxy implementation contributes a row. Train-only standardization and, if needed, randomized PCA precede editing. Each edit retrains a logistic target probe and four registered leakage attackers: logistic regression, an RBF–Nystroem map with logistic regression, a random forest, and a two-layer MLP. A boosted-tree attacker is fitted on the training partition but withheld from edit construction, selection, and certification; it is used only as out-of-portfolio attacker-class stress evidence and is not part of the guarantee.

The official training split is divided into eraser-training and construction; pinned adapters use both as required, while fresh targets and attackers fit only eraser-training. Untouched validation atoms define a finite reference law P. On a disjoint design fold, the rarest feasible supported environment–source–target cell receives ratio Γ, nonfocus atoms receive the unique mass-preserving residual weight, and other environments are unchanged. The induced conditional caps are therefore exact. Receipts record the focus cell, probability-vector hash, profile, and cap check. Certification draws come from P; primary violations and opportunity labels use exact Q-weighted expectations. A separate 50,000-draw replay is diagnostic only. This establishes ambiguity membership for a finite-reference replay, not new patients, comments, images, or recordings.

The primary profile uses $\Gamma = 1.1$, $\delta = 0.05$, and 4,000 observations per dataset–seed. Contracts (τ, λ) are $(0.40, 0.70)$ for Bios, $(0.075, 0.80)$ for CivilComments, $(0.20, 0.55)$ for GaitPDB, and $(0.10, 0.90)$ for Waterbirds, selected by a deterministic seeds 5–12 design grid and not interpreted as normative tolerances. Every cell receives a 15% evidence floor; the remainder follows the design candidate's squared inverse-margin score. Matched analyses use uniform allocation, $\Gamma \in \{1.1, 1.25, 1.5\}$, and $N \in \{1000, 2000, 4000, 8000\}$. Radii are censored at the preregistered cap 8. An earlier analyzer used cap 4; receipts and fixed-profile decisions are unaffected. Cap 8 is authoritative, cap 4 is reported as sensitivity, and an independent cap-8 replay must agree; all differences are reported. A separately prespecified additive allocator uses disjoint certification streams and cannot rescue a failed primary gate.

After that 64-seed primary failed only the usefulness gate, we locked an independent follow-up before any seeds 109–172 outcome files existed. The follow-up keeps the same datasets, eraser frontier, contracts, $\Gamma = 1.1$, 15% evidence floor, and seed-cluster inference, but promotes the previously registered 8,000-observation targeted-allocation sensitivity row to the follow-up primary. The follow-up is not pooled with the first primary and does not retroactively make that failed gate pass; it is a separate post-failure confirmation test with its own sealed replay and first-read comparator.

The disjoint seeds 13–44 design block had 14 favorable, zero adverse, and 18 tied differences, giving smoothed exact-sign power above 0.999 at 64 seeds. Sixty-four also exceeds the 59-seed minimum for a zero-event one-sided 95% upper bound of 0.05.

Nine matched rules share candidates, audits, thresholds, shifts, seeds, and a utility tie-break: always deploy, validation selection, IID LTT, robust point selection, a generic scalar robust certificate, VERA fixed profile, vector envelope, common radius, and an exact shifted-law opportunity oracle. The scalar rule averages threshold-normalized component upper bounds; VERA requires every coordinate to pass. Only eligibility changes, separating eraser quality from deployment decisions and generic robust control.

A prespecified stress compares always deploy, validation selection, and the vector rule at $\kappa \in \{0.75, 1, 1.25\}$, with $\tau_d(\kappa) = \kappa\tau_d$ and $\lambda_d(\kappa) = 0.5 + \kappa(\lambda_d - 0.5)$. All 36 cells are reported. The supporting story requires both naive rules to reach 20% violations somewhere at $\kappa = 1$, the vector rule to remain at most 5% everywhere, and the primary safety/usefulness gates to pass; it cannot rescue the primary decision.

The seed cluster is the independent unit. Efficacy is the within-seed reduction in validation-selection versus vector-envelope violations, tested by a two-sided exact sign test. Safety uses one rotating dataset per seed and a one-sided Clopper–Pearson gate that passes only with zero false acceptances; AM–GM makes this conservative even when seed risks differ. Any event fails the gate. Usefulness is safe retention: a 20,000 whole-seed bootstrap (fixed seed 2027071601) must have a 95% lower bound of at least 20%, and vector retention must be at least twice common-radius retention. Every gate must pass. Per- dataset tests use Holm correction; missing or corrupt receipts are rerun without seed replacement, never excluded by performance.

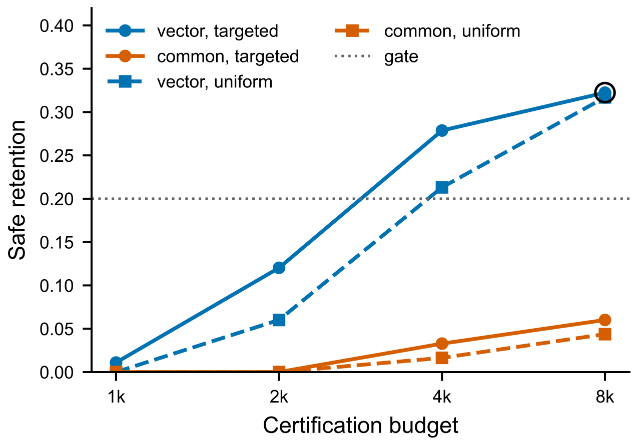

Figure 2: Independent follow-up evidence-size sensitivity at $\Gamma = 1.1$. The circled point is the locked follow-up primary vector rule (budget 8,000, targeted allocation), which clears the registered usefulness gate with zero shifted-contract violations. Lower budgets and uniform allocation are reported as sensitivity analyses.

6 Results

Controlled-shift primary and independent follow-up. The first controlled-shift primary remains a disclosed mixed result: validation selection violated 19/186 deployed decisions, whereas the VERA vector envelope deployed 38 edits with 0 violations; safety, paired reduction, and vector/common advantage passed, but usefulness failed because safe retention was 38/187 (20.3%) with 95% seed-bootstrap CI [15.1%, 25.5%]. The independent post-failure follow-up is not pooled with that result, but it passed as a new locked test: validation selection violated 24/186 deployed decisions, VERA deployed 59 edits with 0 violations, and the paired sign test had 22 favorable seed clusters, 0 adverse, and $p = 4.77e-07$.

Follow-up safety and useful retention. In the independent follow-up, the rotating-sentinel safety gate passed with 0/64 false acceptances and a one-sided 95% Clopper–Pearson upper bound of 4.6%. Across all follow-up VERA vector deployments, shifted-law violations were 0/59. Follow-up safe retention was 59/183 (32.2%) with 95% seed-bootstrap CI [27.5%, 36.8%], clearing the registered 20% lower-bound gate. The common-radius rule retained 11/183 (6.0%); the vector/common ratio was 5.36 with 95% interval [3.5, 10.4], so the vector-advantage gate also passed.

Follow-up allocation and evidence diagnosis. The follow-up evidence-size sweep shows the effect of certification budget: at $\Gamma = 1.1$, the targeted vector rule retained 2/183, 22/183, 51/183, and 59/183 oracle-safe opportunities at budgets 1,000, 2,000, 4,000, and 8,000, respectively, with zero deployed violations in every row. The 8,000-observation row was locked as the independent follow-up primary before seeds 109–172 were run. The follow-up geometry remains narrow but useful: among 59 vector deployments, the median common radius was 1.040, with range [1.002, 1.183]. The most frequent limiting coordinate was target::environment=0 (39 of 59 deployments), followed by source::0 (14 of 59).

Registered stress and negative results. The held-out boosted-tree stress remains outside the formal registered attacker guarantee: it found 5 violations among 59 follow-up vector deployments (8.5% unsafe), so the paper reports it as portfolio-scope stress rather than as a coverage claim. The follow-up did not deploy uniformly across domains: VERA vector abstained on Bios and CivilComments, deployed 14/64 GaitPDB seed decisions and 45/64 Waterbirds seed decisions, and had zero shifted-contract violations on every deployment. This is the intended abstain-or-edit behavior under the supported-envelope contract. The first 64-seed primary's failed usefulness gate remains mandatory negative evidence and is not rescued, pooled, or replaced by the follow-up. The independent follow-up is reported as a separately locked post-failure confirmation at the preregistered higher evidence budget; secondary threshold-stress, held-out-attacker, and dataset-specific rows remain ineligible as primary substitutes.

Prior IID uncertainty-control evidence. In a separate 32-seed study, point selection violated 35/128 contracts (27.3%); the $\Gamma = 1$ certificate violated 1/128 (0.8%) and retained 52/102 safe opportunities (51.0%). All four paired dataset comparisons favored certification after Holm correction. The composite endpoint nevertheless failed because GaitPDB's 5/32 point violations missed its prespecified 20% baseline-severity floor. This supports IID uncertainty control, not $\Gamma > 1$ robustness.

Registered attackers are complementary. In the earlier $\Gamma = 1$ ablation, the full portfolio deployed 12.5% with zero measured violations. Linear-only deployed 43.6% but violated 98/288 (34.0%) complete-portfolio contracts; linear plus RBF violated 17.7%, and dropping MLP produced 43/288 violations. Forest, MLP, RBF, and linear were limiting in 22, 10, 9, and 4 of 45 deployments. This motivates heterogeneous attackers but is not controlled-shift evidence.

Theory tracks controlled data. The exact study had zero false acceptances in 108,000 trials, all 54 abstention rates inside simultaneous prediction bands, and valid coverage in a separate 216-cell grid. These validate the finite-sample implementation; uncontrolled external shifts do not establish ambiguity-set membership.

7 Reproducibility

The anonymous supplement contains code, hashed preregistrations, compact receipts, audit scripts, and manuscript sources. Public-dataset derivatives are represented by manifests and content hashes. Receipts record parent and upstream commits, split hashes, and per-example array hashes. Prior audits validate 1,000 official runs and zero proxy rows. The controlled-shift study adds 1,280 official-method runs and 3,072 closed audit-array archives, all validated by the strict receipt audit before outcome access. A code-path-independent cap-8 replay rehashes arrays, reconstructs paired

Rule	Deployments	Viol./deployed	Safe retention	Median radius
Always deploy	256/256 (100.0%)	128/256 (50.0%)	128/183 (69.9%)	NA
Validation selection	186/256 (72.7%)	24/186 (12.9%)	162/183 (88.5%)	NA
IID LTT	109/256 (42.6%)	0/109 (0.0%)	109/183 (59.6%)	NA
Robust point estimate	98/256 (38.3%)	3/98 (3.1%)	95/183 (51.9%)	NA
Scalar robust certificate	256/256 (100.0%)	126/256 (49.2%)	130/183 (71.0%)	NA
VERA fixed profile	67/256 (26.2%)	0/67 (0.0%)	67/183 (36.6%)	NA
VERA vector envelope	59/256 (23.0%)	0/59 (0.0%)	59/183 (32.2%)	1.04
VERA common radius	11/256 (4.3%)	0/11 (0.0%)	11/183 (6.0%)	NA
External oracle	183/256 (71.5%)	0/183 (0.0%)	183/183 (100.0%)	NA

Table 1: Independent post-failure controlled-shift follow-up at $\Gamma = 1.1$, budget 8,000, and targeted allocation with a 15% evidence floor. Safe retention is retained oracle-safe opportunities over all oracle-safe opportunities. Unlike the first 64-seed primary, this independently locked follow-up passes every registered gate.

harm, verifies shift membership, and recomputes bounds, decisions, envelopes, and summaries; the committed three-way comparator reported zero cap-8 mismatches and zero disallowed cap-4 differences before the scientific first read. The follow-up repeats the same closed-set receipt audit on a fresh 1,280-run seed block and uses a separate sealed independent replay plus protocol cap-8 analyzer; its committed two-way comparator reported zero cap-8 mismatches before the follow-up scientific first read.

Generative-AI assistance disclosure. OpenAI Codex was used extensively for research ideation, literature discovery, theorem and proof drafting, software implementation, experiment orchestration, statistical-analysis code, figure and manuscript drafting, and editing. No AI system is an author or a citable source. The human author or authors are responsible for independently verifying every claim, proof, implementation, reference, data right, and policy obligation before submission.

8 Limitations and Conclusion

VERA assumes bounded audits, certification independent of construction, and deployment support covered by certification. Γ is declared, not estimated. A finite attacker portfolio cannot establish universal erasure, fairness, or future-task safety; paired harm covers only the registered target and learner. Camelyon17 is not clinical validation, and neither medical benchmark supplies clinical deployment evidence. The allocator shares each source-class draw across attackers but samples target-environment and source-class streams separately, so its minimax claim excludes a general joint stratified design. Seeds measure fitting and split variation, not random deployment domains; each seed's four datasets form one cluster. The controlled result concerns a finite reference law with known reweighting, not membership of an unmodeled deployment. The successful 8,000-observation follow-up was locked only after the first 4,000-observation primary missed usefulness, so it is reported as independent post-failure confirmation rather than as an unqualified original-primary success. Every registered outcome is reported whether or not the gates pass.

Responsible use. Concept removal can conceal an attribute without correcting structural bias. A VERA certificate is not privacy, legal fairness, or fitness for high-stakes deployment. Experiments use released secondary benchmarks and no new participants; the release excludes raw biographies, comments, medical images, gait recordings, and participant-linked identifiers.

Within those boundaries, VERA changes the output of representation erasure from “apply this edit” to a falsifiable statement: this edit meets these paired target and attacker contracts up to this supported reweighting budget. When the data cannot justify that sentence, abstention is the result rather than a hidden limitation.

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